When Ā-movement cannot cross CPs: A feature-based account of improper movement

Huacheng Cao[1*]

[1*]Department of Linguistics, Division of Psychology and Language Sciences, UCL, London, UK.

Corresponding author(s). E-mail(s): caohuacheng9704@163.com;

Abstract

The Williams Cycle, which is based on the longstanding idea of the ban on improper movement, encounters a non-trivial problem when we consider the fact that long-distance movement proceeds through clause-internal intermediate positions. This paper makes progress in solving this problem by investigating a sharp locality asymmetry in the Mandarin *lian*. . . *dou* construction: movement to a TP-internal position is clause-bounded (it can span TPs but not CPs), while movement to a TP-external position can cross CP boundaries. Crucially, the "blocked" TP-internal position is still required as an intermediate step in long-distance movement that terminates at the TP-external position. To explain this, I propose a feature-based reformulation of the Williams Cycle. The proposed constraint regulates where features are interpreted (at terminal steps) along the movement path, while ignoring intermediate steps motivated by successive cyclicity. This result not only has repercussions for the proper analytical treatment of improper movement but also sheds light on broader typological interactions between cyclic movement and clause-internal operations.

Keywords: Improper movement, locality, Williams Cycle, Cyclicity, Mandarin Chinese

1 Introduction

Locality mismatches arise when a given structure permits one type of movement to proceed out of it, while simultaneously prohibiting another type. For example, in English, Ā-movement (1a), not A-movement (1b), can cross CP boundaries.

(1) a. Who$_1$ does it seem [$_{CP}$ t$_1$ has left]?

b. * Who$_1$/John$_1$ seems [$_{\text{CP}}$ t$_1$ has left](?)

The standard account of the contrast between (1a–b) involves a conspiracy of two constraints: (i) movement out of a CP must proceed through its specifier, an Ā-position (Chomsky 1973, 1981, 1986); and (ii) a ban on “improper movement”, as stated in (2) (Chomsky 1973, 1981; May 1979).

(2) Ban on Improper Movement (BOIM):
Movement cannot proceed from an Ā-position to an A-position.

Since the seminal work by van Riemsdijk and Williams (1981), a growing body of work has shown that locality mismatches extend beyond the binary distinction between A- and Ā-movement (e.g., Müller and Sternefeld 1993; Grewendorf 2003; Williams 2003, 2011, 2013; Svenonius 2004; Abels 2007, 2012b; Neeleman and van de Koot 2010; Müller 2014a,b; Keine 2019, 2020; Poole 2023). These typological findings motivate a revision of the ban on “improper movement”, which must be generalized to capture the locality properties of different movement types.

One prominent attempt at such a general theory is the *Williams Cycle* (WC) (Williams 2003, 2011, 2013). In Williams (2003), syntax is divided into multiple levels of representations, each associated with a distinct movement type. Since all levels are strictly ordered with respect to each other, specific movement types can only be extended by particular others. One theorem that follows directly from the WC is a generalized version of the BOIM, as in (3), adopted from Poole (2023, p. 348).

(3) Generalized Ban on Improper Movement (GBOIM)
Movement to [Spec, XP] cannot proceed from [Spec, YP] or across YP, where Y is higher than X in the functional sequence (henceforth *fseq*: C > T > v > V).

The WC attributes the locality mismatch in (1a–b) to a general ban on admissible sequences of movement steps: movement cannot proceed from inside a CP to [Spec, TP], as schematized in (4). When long-distance movement reaches the embedded [Spec, CP], the final landing site must be the matrix CP, not the matrix TP, because [Spec, TP] is structurally lower than [Spec, CP] in the *fseq*.

(4) [$_{\text{CP}}$ __✓__ [$_{\text{TP}}$ __✗__ [$_{\text{CP}}$ DP ...

Several reinterpretations of the WC have been proposed by Abels (2007, 2012a,b), Neeleman and van de Koot (2010), Müller (2014a,b), Keine (2019, 2020) and Poole (2023), amongst others. This paper focuses on a general tension between these WC-based accounts of improper movement and the successive-cyclic nature of movement, as already noted by Abels (2007); Müller (2014a); Keine (2019); Poole (2023). To illustrate the problem, consider the schematic representations of A- and Ā- movement in (5) and (6). Under the standard assumption that CPs and vPs constitute phases, both A- and Ā- movement involve an intermediate step from the embedded [Spec, CP] (t$'_1$) to the matrix [Spec, vP] (t$''_1$). Notably, this is the only movement step that proceeds from a higher projection to a lower one in the derivation. To capture the locality mismatch between A- and Ā-movement, this [Spec, CP]-to-[Spec, vP] step should have a different status. It must be well-formed in (5), where the final landing site reaches the same structural height as the embedded [Spec, CP], but ruled out in (6), where it does not. The crucial problem for the GBOIM, however, is that it fails to capture this distinction. If the GBOIM applies to intermediate movement steps, it incorrectly rules out well-formed instances of long-distance Ā-movement such as (5), rendering the grammar overly restrictive. By contrast, if intermediate movement is exempt from the GBOIM, the grammar incorrectly permits illicit A-movement configurations such as (6), making it overly permissive.

(5) $[_{\text{CP}}$ Who_1 $\ldots [_{\text{vP}}$ t_1'' $\ldots [_{\text{CP}}$ t_1' $\ldots \text{t}_1$ $\ldots$

(6) $*[_{\text{TP}}$ John_1 $\ldots [_{\text{vP}}$ t_1'' $\ldots [_{\text{CP}}$ t_1' $\ldots \text{t}_1$ $\ldots$

Two solutions have been proposed in the literature, differing in whether they acknowledge the phasehood of vPs. Müller (2014a,b) preserves vP phases and develops an approach in which the WC becomes operative only at the final landing site. In a nutshell, [Spec, CP]-to-[Spec, vP] movement is obligatory. A movement chain is deemed improper if the extended projection hosting the final landing site is structurally lower than the extended projection containing the launching site in the *fseq*.[1] Keine (2019, 2020) and Poole (2023) argue that this problem disappears if vP is not treated as a phase. On this view, there is no obligatory intermediate movement from [Spec, CP] to [Spec, vP]. The contrast between (5) and (6) is instead explained by the grammaticality of [Spec, CP]-to-[Spec, CP] movement in (5) and the ungrammaticality of [Spec, CP]-to-[Spec, TP] movement in (6).

Drawing on patterns of (im)proper movement in the *lian*... *dou* construction in Mandarin Chinese, I argue that Müller's (2014a, b) approach is on the right track. In the *lian*... *dou* construction, the focus particle *lian* forms a constituent with the focused phrase. The resulting *lian*-phrase must move to a position above *dou*, surfacing either below the subject (TP-internal) or above the subject (TP-external), as in (7). Throughout the paper, I use *TP-internal movement* and *TP-external movement* as convenient descriptive labels for these two movement types.

[1] The idea that the WC applies only at the final landing site is not unique to Müller (2014a,b); it is also implied in the analyses of Abels (2007) and Neeleman and van de Koot (2010). Abels (2007) assumes that movement operations yield chains and proposes a mechanism that constrains subsequent operations targeting the head of chains. Neeleman and van de Koot (2010) interpret the Williams Cycle as a hierarchy of selectional requirements, wherein a superordinate requirement triggers the reconstruction of all subordinate ones. It is conceivable that a selectional requirement triggers reconstruction only when it reaches the position where the requirement is satisfied.

(7) <lian chongzi$_1$ > Zhangsan <lian chongzi$_1$ > dou gan chi t$_1$.
lian insect Zhangsan lian insect dou dare eat
'Zhangsan dares to eat even insects.'

The locality mismatch between TP-internal and TP-external movement tracks the structural height of their landing sites. TP-external movement can cross TPs (8a) and CPs (8b). In contrast, TP-internal movement can proceed only out of TPs (8c–d), even when the focus undergoes Ā-movement.[2] Crucially, the derivation of grammatical TP-external movement in (8b) properly contains the derivation of ungrammatical TP-internal movement in (8d). While the TP-internal position is "blocked" as a final landing site in (8d), it serves as an intermediate landing site in (8b), as indicated by t'_1. This contrast reveals a fundamental property of syntactic locality: a position that imposes locality constraints on terminal movement into it does not impose the same constraints on intermediate movement into it. By contrast, the alternative solution proposed by Keine (2019, 2020) and Poole (2023) is not viable for explaining the pattern in (8). I show that the TP-internal intermediate landing site is not merely a byproduct of clause-internal phases (e.g., vP). Rather, it is empirically supported by reconstruction effects and by the strict local dependency between *dou* and the fronted *lian*-phrase. Therefore, even if vPs are not phases, the empirical puzzle still demands explanation. Specifically, movement from [Spec, CP] to a TP-internal position is well-formed only when it is intermediate rather than terminal.

(8) (Im)proper movement in the *lian*... *dou* construction

a.		lian-XP$_1$	Subject	t'_1	dou ... [$_{\text{TP}}$ ... t$_1$ ...
b.		lian-XP$_1$	Subject	t'_1	dou ... [$_{\text{CP}}$ ... [$_{\text{TP}}$ ... t$_1$ ...
c.			Subject	lian-XP$_1$	dou ... [$_{\text{TP}}$ ... t$_1$ ...
d.	*		Subject	lian-XP$_1$	dou ... [$_{\text{CP}}$ ... [$_{\text{TP}}$ ... t$_1$ ...

Based on the data, I argue that locality constraints on movement must be formulated in a way that they selectively apply to positions that host terminal movement, while ignoring intermediate movement. I follow Abels (2012b) and van Urk (2015) in distinguishing terminal from intermediate movement in terms of feature interpretability. While both terminal and intermediate movement steps are driven by feature valuation, only terminal movement triggers feature interpretation. For expository purposes, terminal movement is referred to as *feature interpretation* steps, and

[2] I assume that embedded clauses differ in structural sizes and their syntactic properties, such as finiteness, are functions of their sizes. Yan and Meadows (2025) provide evidence that finite clauses contain more functional structure than clauses in Mandarin Chinese. Finite clauses permit topicalization within their domain, and the topicalized object can precede sentence-level adverbs like *xingkui* 'fortunately', as in (i). In contrast, non-finite clauses do not allow both, as in (ii). For concreteness, I take finite clauses to be CPs, and non-finite clauses to be defective TPs. The question of how finiteness is encoded in Mandarin Chinese is orthogonal to the present discussion; see Huang (2024) and references therein.

(i) Wassily renwei [$_{\text{CP}}$ [na-ben xiaoshuo]$_1$(-a) xingkui [$_{\text{TP}}$ Piet kan-le san-bian t$_1$]]
Wassily think that-CLF novel TOP fortunately Piet read-ASP three-time
'Wassily thinks that, that novel, fortunately, Piet read three times.'

(ii) * Wassily qiangpo Piet [$_{\text{TP}}$ [na-ben xiaoshuo]$_1$(-a) xingkui kan-le san-bian t$_1$]
Wassily force Piet that-CLF novel TOP fortunately read-ASP three-time
Intended: 'As for that novel, fortunately, Wassily forced Piet to read three times.'

intermediate movement as *feature valuation* steps. I revise the WC as a constraint on admissible sequences of feature interpretation steps along a movement path. A movement chain is deemed improper if a movement step that interprets a given feature (e.g., [$\bar{\text{A}}$] features) is followed by a movement step that interprets certain other features (e.g., [A] features).

The argumentation proceeds as follows. In Sect. 2, I present the core data underlying the pattern schematized in (8). I argue on the basis of these data that the WC must be reformulated as a constraint applying solely to terminal movement, while ignoring intermediate movement steps. Sect. 3 develops this reformulation through a feature-based approach to movement. Sect. 4 introduces a core ingredient for the analysis of the pattern in (8), namely, the constraint on the ordering of features and associated operations in Mandarin Chinese. Sect. 5 shows that the reformulated WC correctly predicts the patterns of cross-clausal movement in the *lian... dou* construction. Sect. 6 shows that the pattern in (8) is not an isolated case. Instead, it instantiates a particular type of interaction between syntactic operations across languages, which provides further support for the reformulation developed in this paper. In Sect. 7, we zoom out to consider an intriguing typological generalization regarding the connection between the locality of a movement type (short vs. long) and the structural height of that movement type's landing site (low vs. high). Sect. 8 concludes.

2 Cross-clausal movement in the *lian... dou* construction

2.1 TP-internal movement

In Mandarin Chinese, the *lian... dou* construction involves focus movement and yields an interpretation comparable to English *even* (e.g., Qu 1994; Shyu 1995; Badan 2008; Constant and Gu 2010; Chen, F. 2023; Cao 2025a). As shown in example (9), *dou* appears in a preverbal position. The particle *lian* combines with the focused object. The *lian*-phrase (i.e., *lian* plus the focused object) moves to a TP-internal position between the subject and *dou*.

(9) Zhangsan lian chongzi$_1$ dou gan chi t$_1$.
Zhangsan lian insect dou dare eat
'Zhangsan dares to eat even insects.'

TP-internal movement of DPs exhibits typical A-properties (Shyu 1995; Chen 2023; Cao 2025a). For example, movement of DP foci creates new antecedents for anaphors at the landing site, as illustrated by the contrast between (10) and (11).

(10) * Wo bi ta-ziji$_1$ de erzi $[_{\text{TP}}$ chumai le Lisi$_1$].
I force s/he-self POSS son betray ASP Lisi
Intended: 'I forced his$_1$ own son to betray Lisi$_1$.'

(11) Wo lian Lisi$_1$ dou bi ta-ziji$_1$ de erzi $[_{\text{TP}}$ chumai le t$_1$].
I lian Lisi dou force s/he-self POSS son betray ASP
'Even Lisi$_1$, I forced his$_1$ own son to betray him$_1$.'

Another A-property that will become important here is that moved DPs do not reconstruct below *dou*.[3] Example (12) is ungrammatical because the anaphor cannot be bound under reconstruction. Scope reconstruction is not possible either. Example (13) allows only the surface scope reading, indicating that the fronted DP does not reconstruct into the scope of the universal *meige xuesheng*.

(12) * Wo lian [$_{\text{DP}}$ ta-ziji$_2$ de fangzi]$_1$ dou bi Li$_2$ [$_{\text{TP}}$ maidiao le t$_1$].
I lian s/he-self POSS house dou force Li sell ASP
Intended: 'I forced Li$_2$ to sell even his$_2$ own house.'

(13) Li lian [yi-qian de jiang-jin]$_1$ dou fengei le meige xuesheng t$_1$.
Li lian one-thousand ADJ bonus-cash dou give ASP every student
$\exists > \forall$: 'Lisi distributed even one thousand yuan (in total) to every student.'
*$\forall > \exists$: 'Zhangsan gave each student even one thousand yuan.'

With respect to locality, TP-internal movement can proceed out of TPs, as in (14), but not out of CPs, as in (15–16). Examples (15–16) show that neither the embedded subject nor the embedded object can move out of the embedded CP to a TP-internal position in the matrix clause. Therefore, the locality constraints on TP-internal movement apply to embedded DPs in general, regardless of the height of their base positions.

(14) Zhang lian [$_{\text{DP}}$ zhe-ben shu]$_1$ dou bi [$_{\text{TP}}$ Wu kan-wan t$_1$].
Zhang lian this-CLF book dou force Wu read-finish]
'Zhang forced Li to ask Wu to read even this book.'

(15) * Lisi lian Zhangsan$_1$ dou juede [$_{\text{CP}}$ t$_1$ ma guo laoban]
Lisi lian Zhangsan dou think scold EXP boss
Intended: 'Even Zhangsan$_1$, Lisi thinks he$_1$ scolded the boss.'

(16) * Lisi lian laobao$_1$ dou juede [$_{\text{CP}}$ Zhangsan ma guo t$_1$]
Lisi lian boss dou think Zhangsan scold EXP
Intended: 'Even the boss$_1$, Lisi thinks Zhangsan scolded him$_1$.'

Previous research attributes the ungrammaticality of (15)–(16) to the clause-boundness of A-movement (Shyu 1995; Constant and Gu 2010; Chen 2023). However, this explanation becomes inadequate once we consider TP-internal movement of VP

[3] In English, A-movement reconstructs only in limited environments, such as raising constructions like (i). (May 1977; Barss 1986; Hornstein 1995; Fox 1999; Lebeaux 2009). By contrast, in Mandarin Chinese, DP fronting to a preverbal position generally does not reconstruct, as in (ii) (Qu 1994; Ernst and Wang 1995; Shyu 2001; Sybesma 2021; Cao 2025a). The cross-linguistic variation in reconstruction in A-movement is a separate issue, which I leave for future work. Nevertheless, the overarching A/Ā distinction in reconstruction remains valid in Mandarin Chinese: reconstruction under A-movement is more limited than under Ā-movement. As I will show later, Ā-movement in Mandarin Chinese does show reconstruction effects.

(i) (Lebeaux 2009, p. 5)
Two women seem to be expected to dance with every senator.
Two $\gg \forall$: There are exactly two women who dance with every senator.
$\forall \gg$ two: Every senator has two different women who dance with them.

(ii) Li yuanyi [yi-qian-kuai de jiang-jin]$_1$ fengei meige xuesheng t$_1$.
Li willing one-thousand-yuan ADJ bonus-cash give every student
$\exists > \forall$: 'Li is willing to distribute one thousand yuan (in total) to every student.'
*$\forall > \exists$: 'Li is willing to give each student even one thousand yuan.'

foci. Cao (2025a) shows that VP foci undergo $\bar{\text{A}}$-movement to a TP-internal position. In sharp contrast to movement of DPs, movement of VPs reconstructs for anaphor binding (17) and scope (18). If clause-boundedness were a property of A-movement per se, VP foci—which undergo $\bar{\text{A}}$-movement—should not be subject to the same locality restriction. We would then predict that VP foci could undergo long-distance movement to a TP-internal position in the matrix clause. However, this prediction is not borne out by (19).

(17) Wo lian [$_{\text{VP}}$ maidiao ta-ziji$_2$ de fangzi]$_1$ dou bi Li$_2$ [$_{\text{TP}}$ zuo le t$_1$].
I lian sell s/he-self POSS house dou force Li do ASP
'I even forced Li$_2$ to sell his$_2$ own house.'

(18) Lisi$_1$ lian [$_{\text{VP}}$ gei ta$_1$ yi-bai-kuai]$_2$ dou bi mei-ge haizi zuo t$_2$.
Lisi lian give s/he one-hundred-yuan dou force every-CLF child do
$\forall > \exists$: 'Lisi$_1$ forced each student even to give him$_1$ one hundred yuan.'
*$\exists > \forall$: 'Lisi$_1$ forced every child even to give him$_1$ one hundred yuan (in total).'

(19) * Zhangsan lian [$_{\text{VP}}$ chi chongzi]$_1$ dou xiangxin [$_{\text{CP}}$ Lisi gan zuo t$_1$].
Zhangsan lian eat insect dou believe Lisi dare do
Intended: 'Even eat insects, Zhangsan believes that Lisi dares to do.'

The locality profile of TP-internal movement is summarized in (20). Crucially, TP-internal movement is sensitive to the size of the complement clause from which the focus moves, regardless of whether the movement is A- or $\bar{\text{A}}$-movement.

(20) a. Subject lian-XP$_1$ dou … [$_{\text{TP}}$ … t$_1$ …
b. * Subject lian-XP$_1$ dou … [$_{\text{CP}}$ … [$_{\text{TP}}$ … t$_1$ …

2.2 TP-external movement

(21) Lian [$_{\text{DP}}$ zhe-ben shu]$_1$ wo dou bi Zhang [$_{\text{TP}}$ kan-wan le t$_1$].
lian this-CLF book I dou force Zhang read-finish ASP
'I forced Zhang to read even this book.'

(22) Lian [$_{\text{VP}}$ kan-wan zhe-ben shu]$_1$ wo dou bi Zhang [$_{\text{TP}}$ zuo le t$_1$].
lian read-finish this-CLF book I dou force Zhang do ASP
'I forced Zhang to even read this book.'

(23) Lian laobao$_1$ Lisi dou juede [$_{\text{CP}}$ Zhangsan ma guo t$_1$]
lian boss Lisi dou think Zhangsan scold EXP
'Even the boss$_1$, Lisi thinks Zhangsan scolded him$_1$.'

(24) Lian Zhangsan$_1$ Lisi dou juede [$_{\text{CP}}$ t$_1$ ma guo laoban]
lian Zhangsan Lisi dou think scold EXP boss
'Even Zhangsan$_1$, Lisi thinks he$_1$ scolded the boss.'

(25) Lian [$_{\text{VP}}$ chi chongzi]$_1$ Zhangsan dou xiangxin [$_{\text{CP}}$ Lisi gan zuo t$_1$].
lian eat insect Zhangsan dou believe Lisi dare do
'Even eat insects, Zhangsan believes that Lisi dares to do.'

In contrast, TP-external movement is insensitive to the size of the complement clause from which the focus moves. It can proceed out of both TPs (21–22) and CPs (23–25). Examples (23–24) show that both the embedded subject and object can move long-distance to a TP-external position. Long-distance TP-external movement is also permitted when the focus is a VP, as in (25).[4]

The full pattern of cross-clausal movement in the *lian*. . . *dou* construction is summarized in (26).

(26) (Im)proper movement in the *lian*. . . *dou* construction

a.		lian-XP_1	Subject	t'_1	dou . . . [$_{TP}$ . . . t_1 . . .
b.		lian-XP_1	Subject	t'_1	dou . . . [$_{CP}$ . . . [$_{TP}$ . . . t_1 . . .
c.			Subject	lian-XP_1	dou . . . [$_{TP}$ . . . t_1 . . .
d.	*		Subject	lian-XP_1	dou . . . [$_{CP}$ . . . [$_{TP}$ . . . t_1 . . .

2.3 Movement out of double embedding structures

The crucial data point from previous sections is that, in the *lian*. . . *dou* construction, there is a systematic connection between the structural height of a movement type's landing site determines the range of domains from which that movement type may proceed. In other words, the higher the landing site of a movement type, the broader the range of structures that movement type can span.

Further evidence for this interaction comes from movement out of doubly embedded structures. Example (27) illustrates the baseline configuration, in which a CP is embedded within a TP, which in turn is embedded in a matrix clause. Example (28) shows that the object of the most deeply embedded CP cannot move to a position below the matrix subject; only a TP-external position is available.Example (29) differs from (28) in that the fronted object originates in a TP. In this case, movement can target a TP-internal position in the matrix clause. In (28) and (29), the clause containing the base position of the fronted object is separated from the matrix clause by

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[4]Shyu (1995) argues that TP-external *lian*-phrases are base-generated in the left periphery. However, this base-generation analysis is untenable. First, if base generation were possible for long-distance dependencies, it is unclear why *lian*-phrases cannot occur below the matrix subject. Second, as I will show later, TP-external *lian*-phrases reconstruct below the matrix subject. Third, Shyu's main argument is that TP-external *lian*-phrases can fronted out of an indirect question, as in (i). However, indirect questions in Mandarin Chinese do not generally block movement. As shown in (ii), ex-situ cleft constructions—derived by Ā-movement (see Pan 2017 and references therein)—can also span indirect questions. By contrast, neither *lian*-phrases nor clefted phrases can be extracted from relative clauses, which form an absolute barrier to movement, as shown in (iii)–(iv).

(i) Lian zhe-ben shu_1 Li dou xiang zhidao [$_{CP}$ shei yijing kan-wan le t_1].
lian this-CLF book Li dou want know who already read-finish ASP
'Li wants to know who read even this book.'

(ii) Shi zhe-ben shu_1 Li xiang zhidao [$_{CP}$ shei yijing kan-wan le t_1].
COP this-CLF book Li want know who already read-finish ASP
'This is the book that Li wants to know who has read.'

(iii) * Lian $Qiaomusiji_1$ Li dou juede [$_{CP}$ Zhang bu xihuan [$_{DP}$ [t_1 xie de] shu]]
lian Chomsky Li dou think Zhang NEG like write REL book
Intended: 'Li thinks Zhang does not like the book that even Chomsky wrote.'

(iv) * Shi $Qiaomusiji_1$ Li juede [$_{CP}$ Zhang bu xihuan [$_{DP}$ [t_1 xie de] shu]]
COP Chomsky Li think Zhang NEG like write REL book
Intended: 'Li thinks Zhang does not like the book that Chomsky wrote.'

a TP. The two sentences differ only in the size of the clause containing the base position of the moved DP: CP in (28) versus TP in (29). This suggests that TP-internal movement cannot in general apply across CPs, regardless of the depth at which those CPs are embedded.

(27) Li$_1$ dasuan [$_{\text{TP}}$ chengren [$_{\text{CP}}$ ta$_1$ ma guo laoshi]].
Li plan admit s/he scold EXP teacher
'Li$_1$ planned to admit that he$_1$ scolded the teacher.'

(28) <Lian laoshi$_2$ > Li$_1$ <*lian laoshi$_2$ > dou dasuan [$_{\text{TP}}$ chengren [$_{\text{CP}}$ ta$_1$ ma guo t$_2$]].
lian teacher Li lian teacher dou plan admit s/he scold EXP
'Even the teacher$_2$, Li$_1$ planned to admit that he$_1$ scolded him$_2$.'

(29) <Lian laoshi$_2$ > Li$_1$ <Lian laoshi$_2$ > dou dasuan [$_{\text{TP}}$ jixu [$_{\text{TP}}$ ma t$_2$]].
lian teacher Li lian teacher dou plan continue scold
'Even the teacher$_2$, Li$_1$ planned to continue scolding him$_2$.'

2.4 Intermediate movement in the matrix clause

TP-external *lian*-phrases consistently move through a TP-internal position within the matrix clause, as indicated by t'_1 in (30b).

(30) a. Subject lian-XP$_1$ dou ... (*[$_{\text{CP}}$ ...) [$_{\text{TP}}$ ... t$_1$ ...
b. lian-XP$_1$ Subject t'_1 dou ... ([$_{\text{CP}}$ ...) [$_{\text{TP}}$ ... t$_1$ ...

The first piece of evidence concerns the local dependency between *dou* and the *lian*-phrase. In TP-internal movement, the fronted *lian*-phrase must immediately c-command *dou* (Zhang 1997; Cao 2025b).[5] I hereafter refer to this constraint as the immediate c-command condition. This condition rules out examples of DP fronting in (31) and VP fronting in (32), in which the adverb *jingchang* 'often' may attach either above the *lian*-phrase or below *dou*, but not in the position between them.

(31) Mali <jingchang> lian fan$_1$ <*jingchang> dou <jingchang> bu chi t$_1$.
Mary often lian meal often dou often NEG eat
'Mary often does not even eat.'

(32) Wu <jingchang> lian [$_{\text{VP}}$ chi sheng-fan]$_1$ <*jingchang> dou <jingchang> bi Lisi duo t$_1$.
Wu often lian eat left-food often dou often force Lisi do
'Wu often forces Lisi even to eat leftovers.'

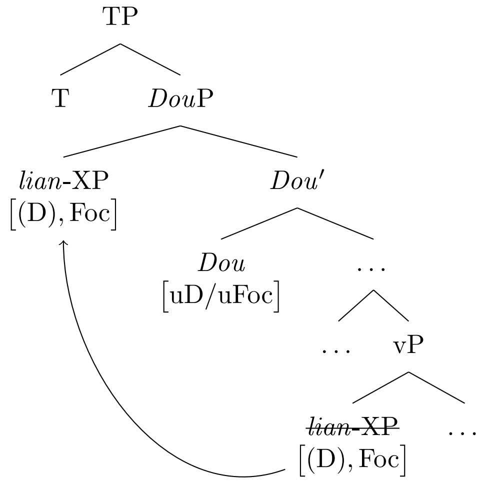
Following Chen (2023) and Cao (2025a), I assume that focus movement is driven by a composite [D/Foc(us)] probe on the head *dou*, as depicted in (33).[6] Under the

[5] A constituent α immediately c-commands β iff there is no γ such that α asymmetrically c-commands γ and γ asymmetrically c-commands β. α asymmetrically c-commands β iff α c-commands β and β does not c-command α.

[6] For discussion of how the composite probe triggers A-movement of DP foci versus Ā-movement of VP foci, see Sect. 3 and Cao (2025a).

standard assumption that feature valuation in movement depends on a specifier–head relation between a probe and its goal, the immediate c-command condition in TP-internal movement follows as a natural consequence.

(33)



In examples (34) and (35), the TP-external *lian*-phrase is not linearly adjacent to *dou*. However, these are not true counterexamples, as the immediate c-command condition is satisfied underlyingly. Given that the head *dou* also appears in the TP-external movement and that the same focus interpretation is available, the same features must be valued in the same configuration in (33). I argue that, in these examples, the *lian*-phrase immediately c-commands *dou* at the TP-internal intermediate landing site t'_1, which required for valuation of the features involved in the TP-external movement. Otherwise, the derivation would crash, as the features on *dou* would remain unvalued.

(34) lian xizhuren_1 Li t'_1 dou ma guo t_1.
lian dean Li dou scold EXP
'Li even scolded the dean.'

(35) lian $[_{\text{VP}}$ chi guoqi-de shiwu $]_1$ Li t'_1 dou bi Wu zuo guo t_1.
lian eat expired-ADJ food Li dou force Wu do EXP
"Li forced Wu to even eat the expired food.'

Further support comes from reconstruction effects in TP-external movement. As with TP-internal movement, TP-external movement of DPs does not reconstruct below *dou*. However, TP-external *lian*-DPs do exhibit reconstruction in the domain above *dou*. In example (36), the anaphor *ta-ziji* can only be bound by the subject *Zhang*, not by the lower argument *Li*. I assume that the binding relation is determined precisely at the TP-internal position t'_3, where the features on the head *dou* are valued.

(36) Lian $[_{\text{DP}}$ $\text{ta-ziji}_{1/*2}$ de pengyou $]_3$ Zhang_1 t'_3 dou bi Li_2 $[_{\text{TP}}$ ma guo t_3].
lian s/he-self POSS friend Zhang dou force Li scold EXP
'Even $\text{his}_{1/*2}$ own friends, Zhang_1 forced Li_2 to scold (them).'

Examples (37) and (38) illustrate long-distance TP-external movement of DP foci. Notably, the binding possibilities of the anaphor track the position of *dou*, as highligted by boldface.[7] In example (37), where *dou* occurs in the matrix clause, the anaphor can only be bound by the matrix subject.[8] By contrast, when *dou* appears in the embedded CP in (38), the anaphor can be bound by either the matrix subject or the embedded subject. This contrast is best explained if the moved DP reconstructs to the TP-internal position above *dou*, but not to any position below it.

(37) Lian [$_{\text{DP}}$ ta-ziji$_{1/*2}$ de pengyou]$_3$ Wu$_1$ t'_3 **dou** juede [$_{\text{CP}}$ Li$_2$ ma guo t_3].
lian s/he-self POSS friend Wu dou think Li scold EXP
'Even his$_{1/*2}$ own friends, Wu$_1$ thinks Li$_2$ scolded (them).'

(38) Lian [$_{\text{DP}}$ ta-ziji$_{1/2}$ de pengyou]$_3$ Wu$_1$ juede [$_{\text{CP}}$ Li$_2$ t'_3 **dou** ma guo t_3].
lian s/he-self POSS friend Wu think Li dou scold EXP
'Even his$_{1/2}$ own friends, Wu$_1$ thinks Li$_2$ scolded (them).'

The existence of this intermediate landing site has important implications for accounts of improper movement. First, it rules out the WC and its variants, such as those proposed by Keine (2019) and Poole (2023). The WC dictates that, once a moving element has reached the embedded [Spec, CP], the next landing site must be the matrix [Spec, CP], skipping all positions within the matrix clause. However, this prediction is falsified, since TP-external movement necessarily involves intermediate landing sites within the matrix clause, even when the matrix clause selects a CP. Second, this intermediate position reveals the core analytical intuition that any adequate account of improper movement must incorporate: a position that imposes locality constraints on terminal movement into it does not impose the same constraints on intermediate movement into it. In the following, I develop such an account within a feature-based theory of movement and show how it captures the Mandarin facts.

3 A feature-based reformulation of the WC

I adopt Abels' (2012b) local derivational approach to movement. This framework models the movement system with a division of labor across two components. First, it assumes a feature-based theory of cyclicity and phases (see also Chomsky 1995, 2007, 2008; Georgi 2014; van Urk 2015). Second, it posits a hierarchy of features and their associated operations (see also Müller and Sternefeld 1993; Grewendorf 2003; Neeleman and van de Koot 2010). Under the feature-based theory of cyclicity and phases, the locality of a movement type follows from how movement-type-specific features are distributed across phase heads (see also Rizzi 1997, 2004a; Legate 2011;

[7] Different positions of *dou* correlate with different scopes of the focus interpretation, see Shyu (2018) for discussion.

[8] One may wonder why reconstruction below *dou* is blocked. I assume that this follows from the properties of the features valued at this position. In cases of *lian*-DPs, the valued [D] feature prevents reconstruction below *dou*. By contrast, VP foci agree with the [D/Foc] probe only in the [Foc] feature. As expected, TP-external *lian*-VPs exhibit reconstruction to their base positions, as in (i).

(i) Lian [$_{\text{VP}}$ ma ta-ziji$_{*1/2}$ de pengyou]$_3$ Zhang$_1$ t'_3 <dou> juede [$_{\text{CP}}$ Li$_2$ <dou> zuo guo t_3].
lian scold s/he-self POSS friend Zhang dou think Li dou do EXP
'Even scold his$_{*1/2}$ own friends, Zhang$_1$ thinks Li$_2$ did.'

Obata and Epstein 2011; Abels 2012a,b; van Urk 2015). The feature hierarchy, in turn, constrains interactions among movement types. For example, a movement chain created by a feature [τ] cannot be extended by movement driven by a feature [σ] if [τ] ranks higher than [σ] in the hierarchy.

3.1 Successive cyclic movement

I assume a derivational model of syntax in which syntactic structure is built from bottom to top by applications of Merge and Agree, and only local domains of the structure are accessible at any point in the derivation (Chomsky 1993, 1995, 2001).

Features participating in Agree are specified for their morphosyntactic and interpretive properties, and are classified into [A] features (e.g., [D] and [ϕ]) and [Ā] features (e.g., [Wh] and [Topic]) (Rizzi 1997; Abels 2012a,b; van Urk 2015). These features can be either valued ([F]) or unvalued ([uF]). Unvalued features must acquire a value during the derivation; otherwise, they are illicit objects at the interface. I adopt the *feature-sharing* implementation of Agree proposed by Pesetsky and Torrego (2007) (see also Frampton and Gutmann 2000; Legate 2005; Bobaljik 2008). In this system, when two features are shared, valuation occurs as long as one of them is already valued. By adopting the feature-sharing system, we gain the technical ability of treating valuation and interpretability as independent notions. This distinction is crucial for distinguishing terminal from intermediate movement steps in the discussion that follows.

Movement is modeled as Internal Merge, that is, the combination of syntactic objects that stand in a part–whole relation. Merge—and thus movement— does not apply freely but is triggered only as a last resort, as formalized in (39).

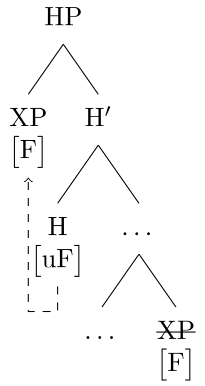
(39) Last Resort (Abels 2012b, p. 105):
A constituent α may only be merged—internally or externally—if that leads to the immediate valuation of a feature.

Following standard approaches to agreement, I assume that feature valuation depends on a syntactic dependency between a head bearing an unvalued feature (probe) and a syntactic object bearing a matching valued feature (goal). I assume that all phrasal movement results from the successive application of Merge and Agree.[9] Specifically, probes that require upward valuation (the goal c-commands the probe) may trigger movement of the closest goal to its specifier, as depicted in (40), the dashed line indicates the upward feature-sharing relation.[10]

[9]The main arguments in this paper are compatible with the alternative view that Agree applies before Merge. On this view, movement is triggered by probes that require downward valuation and also bear a secondary structure-building property in the sense of Heck and Müller (2007). I remain agnostic about the directionality of Agree, as pursuing the issue would take us too far afield. For discussion, see Bjorkman and Zeijlstra (2019) and references therein.

[10]A constituent β is closer to α than γ if α asymmetrically c-commands both β and γ, and β asymmetrically c-commands γ.

(40) Internal merge (movement) of XP



That movement proceeds in a successive cyclic fashion is conceptually motivated by Chomsky (1973, 1977) and has received a wide range of empirical evidence since then (for recent overviews, see Abels 2012b; van Urk 2020). I implement successive cyclicity with the *Phase Impenetrability Condition*, as in (41). The PIC dictates that a goal, which is to value a probe on a head, must move through the specifiers of phases that intervene between the base position of the goal and the specifier of that head. I assume that CPs and vPs are phases (Chomsky 2000, 2001) and that the distribution of probes over phase heads is arbitrary. In other words, Universal Grammar allows phase heads to bear any probes.[11]

(41) Phase Impenetrability Condition (PIC) (Chomsky 2000, p. 108)
In a phase α with the head H, the domain of H is not accessible to operations outside α; only H and its edge are accessible to such operations.

For illustration, consider the derivation of successive-cyclic *wh*-movement in English, as in (42). The notation "$\text{t}_{[\text{F}]}$" (with primes) marks intermediate landing sites for movement triggered by [F]. This notation has no theoretical status in my proposal, serving only as a mnemonic for intermediate positions. Each step is licensed by Last Resort because it results in valuing a [uWh] probe. In (43b, d, f, and h), this is represented as follows. The *u* in [uWh] on C and v is deleted once these probes are valued, after the *wh*-phrase moves to their specifiers.

(42) What does John think Mary will buy?

a. $[_{\text{v}'}\ \text{v}_{[\text{uWh}]}\ [\ \ldots \text{what}_{[\text{Wh}]}\ \ldots$
b. $[_{\text{vP}}\ \text{what}_{[\text{Wh}]}\ [_{\text{v}'}\ \text{v}_{[\text{Wh}]}\ [\ \ldots$
c. $[_{\text{C}'}\ \text{C}_{[\text{uWh}]}\ [_{\text{vP}}\ \text{what}_{[\text{Wh}]}\ [_{\text{vP}}\ \text{v}_{[\text{Wh}]}\ \ldots$
d. $[_{\text{CP}}\ \text{what}_{[\text{Wh}]}\ [_{\text{C}'}\ \text{C}_{[\text{Wh}]}\ [_{\text{vP}}\ \text{t}'_{[\text{Wh}]}\ [_{\text{v}'}\ \text{v}_{[\text{Wh}]}\ \ldots$
e. $[_{\text{v}'}\ \text{v}_{[\text{uWh}]}\ [_{\text{CP}}\ \text{what}_{[\text{Wh}]}\ [_{\text{C}'}\ \text{C}_{[\text{Wh}]}\ [_{\text{vP}}\ \text{t}_{[\text{Wh}]}\ [_{\text{v}'}\ \text{v}_{[\text{Wh}]}\ \ldots$
f. $[_{\text{vP}}\ \text{what}_{[\text{Wh}]}\ [_{\text{v}'}\ \text{v}_{[\text{Wh}]}\ [_{\text{CP}}\ \text{t}''_{[\text{Wh}]}\ [_{\text{C}'}\ \text{C}_{[\text{Wh}]}\ [_{\text{vP}}\ \text{t}'_{[\text{Wh}]}\ [_{\text{v}'}\ \text{v}_{[\text{Wh}]}\ \ldots$

[11] Note that probes are not restricted to phase heads: non-phase heads may also bear probes (e.g., [ϕ] probes on T).

g. $[_{C'}\ C_{[uWh]}\ [_{vP}\ \text{what}_{[Wh]}\ [_{v'}\ v_{[Wh]}\ [_{CP}\ t_{[Wh]}\ [_{C'}\ C_{[Wh]}\ [_{vP}\ t_{[Wh]}\ [_{v'}\ v_{[Wh]} \ldots$

h. $[_{CP}\ \text{what}_{[Wh]}\ [_{C'}\ C_{[Wh]}\ [_{vP}\ t'''_{[Wh]}\ [_{v'}\ v_{[Wh]}\ [_{CP}\ t''_{[Wh]}\ [_{C'}\ C_{[Wh]}\ [_{vP}\ t'_{[Wh]}\ [_{v'}\ v_{[Wh]} \ldots$

Note that the model not only allows successive-cyclic movement, but also enforces it. The derivation will crash if any movement step is skipped or not licensed by the Last Resort condition. Suppose that in (42d), the *wh*-phrase moves directly from its base position to the [Spec, CP]. (43a) shows one possible derivation where the lower v head does not bear a [uWh] feature. This structure is ruled out by the PIC. Before moving to the embedded [Spec, CP], the *wh*-phrase is trapped in the domain of the lower v head, making it inaccessible to the [uWh] probe on C. (43b) illustrates another derivation in which the lower v head bears a [uWh] feature. This structure does not converge for two reasons. First, just like (43a), it violates the PIC. Second, the [uWh] feature on v remains unvalued, resulting in an illicit object at the interface.

(43) Violations of the Phase Impenetrability Condition

a. $^*[_{CP}\ \text{what}_{[Wh]}\ [_{C'}\ C_{[Wh]}\ [_{vP}\ v \ldots$

b. $^*[_{CP}\ \text{what}_{[Wh]}\ [_{C'}\ C_{[Wh]}\ [_{vP}\ v_{[uWh]} \ldots$

As shown in (44), the *wh*-phrase cannot move to [Spec, vP] if the v head does not bear a [uWh] feature. This is because intermediate movement to [Spec, vP] violated the Last Resort condition.

(44) Violations of Last Resort at vP

$^*[_{CP}\ \text{what}_{[Wh]}\ [_{C'}\ C_{[Wh]}\ [_{vP}\ t'_{[Wh]}\ [_{v'}\ v \ldots$

3.2 Terminal versus intermediate movement

Terminal movement is usually associated with morphosyntactic effects (e.g., ϕ-agreement and case assignment) or interpretive effects (e.g., scope taking and information-structure properties) that are unavailable at intermediate movement steps. Therefore, virtually all approaches must distinguish terminal from intermediate movement in one way or another (for a recent overview, see van Urk 2020). In the feature-based approach I adopt here, all movement steps are triggered by designated features but are distinguished by feature interpretability (Chomsky 2000, 2001; Abels 2012b; Georgi 2014; van Urk 2015).[12] Specifically, although both terminal and intermediate movement are driven by feature valuation, only terminal movement feeds interpretation. For expository purposes, I refer to terminal movement as *feature interpretation* steps and intermediate movement as *feature valuation* steps.

[12] I assume that the interpretability of features is a relative notion. A feature may be interpretable with respect to one of the two (or perhaps both) interfaces: the C-I interface and the SM interface. The C-I interface can interpret features that have semantic import, such as features involving tense and [ϕ] features. The SM interface, on the other hand, can interpret features that are relevant for articulation. For a discussion on the interpretability of features, see Pesetsky and Torrego (2007).

Note that feature interpretation steps should not be equated with final landing sites.[13] Final landing sites refer to the surface position of the moved element. By contrast, feature interpretation steps are defined in terms of the interpretation of specific features. Consequently, while final landing sites alway involves feature interpretation, not all feature interpretation steps correspond to final landing sites. For example, a single movement chain may contain multiple feature interpretation steps, each corresponding to the interpretation of specific features along the path. However, the final landing site is the unique position where the moving element is pronounced. In the following, I outline the mechanism of feature interpretation. A full discussion of possible sequences of feature interpretation steps is deferred to the next section.

The core component is a transitive, asymmetric, possibly total order of features and associated operations, as in (45) (Abels 2007). For features $[\sigma] \ll [\tau]$, once the valuation of $[\tau]$ drives movement of a constituent α, the feature $[\sigma]$ contained within α cannot be valued and trigger subsequent movement of α.

(45) The Universal Constraint on the Ordering of Operations in Language (UCOOL) (adapted from Abels 2007, p. 60)
Movement operations are hierarchically ordered, with a feature to the left of $\ll$ ranked lower than features to the right: $[\theta] \ll [\phi] \ll [\bar{\mathrm{A}}]$[14]

To facilitate the following discussion, I label movement steps in successive-cyclic movement in terms of their triggers and landing positions.

(46) A movement step is defined as $\mathrm{M}_{\beta}^{[\mathrm{F}]}$ iff [F] is the trigger for moving a syntactic object α into the specifier of a head β, with β bearing a [uF] probe.

When a probing head carries one feature, and this feature is valued as the result of movement, it automatically serves as the trigger. However, there are also situations in which a movement step results in the valuation of several features. In these cases, the lowest-ranked feature serves as the trigger, and the remaining features will be valued as free riders. For illustration, consider the hypothetical movement chain of ZP in (47), which involved three distinct movement steps. Suppose that $[\sigma]$ ranks lower than $[\tau]$ on UCOOL. $\mathrm{M}_{\mathrm{W}}^{[\sigma]}$ and $\mathrm{M}_{\mathrm{X}}^{[\sigma]}$ are triggered by $[\sigma]$, with $[\tau]$ valued as a free rider. The former step targets [Spec, WP], whereas the latter targets [Spec, XP]. The last step $\mathrm{M}_{\mathrm{Y}}^{[\tau]}$ is triggered by $[\tau]$, targeting [Spec, YP].

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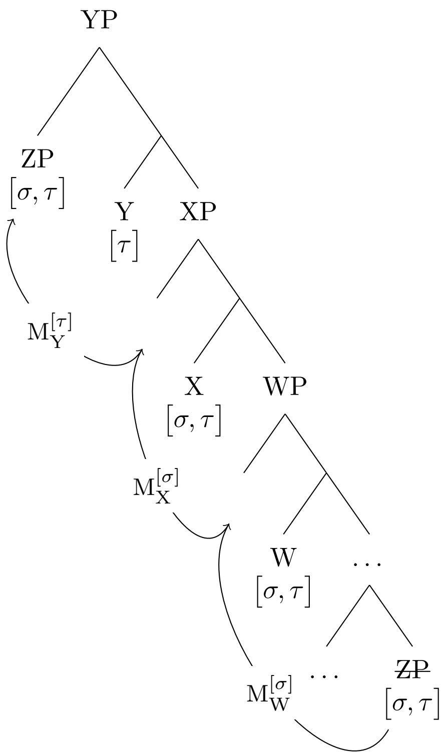
[13] Feature interpretation steps cannot be reduced to criterial movement steps in the sense of Rizzi (2006) either. In a framework assuming criterial movement, once a phrase reaches the criterial position, it cannot undergo further movement. However, extensions of movement chains should not be entirely ruled out, but rather restricted, which is precisely one of the phenomena that the WC is designed to capture.

[14] $[\bar{\mathrm{A}}]$ represents a class of specific Ā-features. Alternatively, $[\bar{\mathrm{A}}]$ may be understood as a flat $[\bar{\mathrm{A}}]$ probe, in the sense of Abels (2012a), which can be satisfied by any Ā-feature present on the goal.

(47)



With the notion of *trigger* clarified for our purposes, we can move on to the mechanism of feature interpretation proposed by Abels (2012b). Valued features must be interpreted, and they are interpreted once under valuation. In other words, while a feature may serve as the trigger for multiple movement steps, only one interpretation step occurs per feature. As a premise, the *height* of probes is defined in terms of asymmetric c-command relations, as in (48). Following Abels (2012b), I assume that the a valued feature is interpreted in the highest position in which it served as the trigger, as shown in (49a-b). Conditions in (49c-d) further prevent the interpretation of the same feature elsewhere along the movement path. In (47), the movement chain involves two interpretation steps. While both $\mathrm{M}_{\mathrm{W}}^{[\sigma]}$ and $\mathrm{M}_{\mathrm{X}}^{[\sigma]}$ are triggered by [σ], $\mathrm{M}_{\mathrm{X}}^{[\sigma]}$ is the interpretation step for [σ] as this step targets the highest probe that bears [uσ]. Meanwhile, [τ] is interpreted at the final landing site, [Spec, YP].

(48) An element α in a syntactic object γ is higher than another element β in γ if and only if α asymmetrically c-commands β.

(49) Principle of interpretation of triggering features:
A movement step $\mathrm{M}_{\beta}^{[\mathrm{F}]}$ of α in γ is the interpretation step for [F] if and only if,
a. it leads to the valuation of [F] between a head β and its moved specifier, and

b. movement of α to the specifier of a higher head δ does not lead to the valuation of [F].

If so, then

a. [F] is interpreted on head β,

b. for all movement steps $\mathrm{M}_{\delta}^{[\mathrm{F}]}$ different from $\mathrm{M}_{\beta}^{[\mathrm{F}]}$, [F] is not interpreted on the corresponding head δ.

The trigger and interpretation of composite probes require further elaboration. Regarding the trigger, the cross-linguistic signature of composite A/Ā constructions provides a crucial clue. Lohninger (2025) shows that composite [A/Ā] probes yield either an A- or an Ā-chain, but never chains with mixed A- and Ā-properties. I argue that understanding this pattern hinges on the ranking of features within the probe. Under the UCOOL, the A/Ā distinction is relativized between any two specific features. Assuming that the triggering feature is the sole determinant of the properties of the movement chain (van Urk 2015), the cross-linguistic signature of composite A/Ā constructions is expected if the true trigger in composite [A/Ā] probes is the lowest-ranked feature involved. Therefore, I propose that, just like bare probes, only the lowest-ranked feature—never free riders—serves as the trigger. That is, when movement results in the valuation of multiple features (either bundled or isolated), it is always the lowest feature that serves as the trigger (see also Lohninger 2025).

The critical role of feature ranking becomes evident in the *lian... dou* construction. Specifically, DP foci are fully matching goals for the [uD/uFoc] probe. Despite both [uD] and [uFoc] being valued, movement of DP foci exhibits A-properties, determined solely by the [D] feature. In contrast, in cases of VP foci, only the [Foc] feature is valued between the probe and the goal, resulting in Ā-movement of VPs.[15] This pattern can be explained straightforwardly if, in Mandarin Chinese, [D] ranks lower than [Foc] (i.e., [D] $\ll$ [Foc]). In the next section, I will show that this is indeed the case.

Now we turn to the interpretation of composite probes. Previous researchers argue that features in composite probes act in unison (e.g., Coon and Bale 2014; van Urk 2015; Bossi and Diercks 2019; Scott 2021; Branan and Erlewine 2024; Lohninger 2025). Under the distinction between feature interpretation and valuation assumed here, I attribute this "act in unison" property to the interpretation of composite probes: all features bundled within a composite probe are interpreted jointly. In movement driven by bare probes, features are never interpreted in their non-triggering steps. However, if movement is driven by composite probes, non-triggering features are also interpreted, even though only the lowest-ranked feature triggers movement. For example, in the *lian... dou* construction, while the [D] feature triggers A-movement of DP foci, the non-triggering [Foc] feature is simultaneously interpreted. I propose a principle of interpretation for composite probes, as in (50). It dictates that a composite probe is interpreted jointly at the highest position where its lowest feature acts as the trigger.

(50) Principle of interpretation of composite probes:
For a composite probe $[\sigma/\tau]$, with $\sigma \ll \tau$, all bundled features are interpreted jointly at the interpretation step of $[\sigma]$. The interpretation of $[\sigma]$ follows the principle of interpretation of triggering features in (49).

[15] I assume that in cases of VP foci, the [D] feature is deleted in the sense of Preminger (2011, 2014).

3.3 Generalized Ban on Improper Feature Interpretation

Recall the crucial analytical generalization we learned from (im)proper movement in the *lian...dou* construction: a position that imposes locality constraints on terminal movement into it does not impose the same constraints on intermediate movement into it. Against this background, the GBOIM under the WC, as repeated here in (51), must be revised because it imposes the constraint unselectively on all movement steps. In other words, a weaker version of the GBOIM is needed.

(51) Generalized Ban on Improper Movement (GBOIM)
Movement to [Spec, XP] cannot proceed from [Spec, YP] or across YP, where Y is higher than X in the functional sequence (henceforth *fseq*: C > T > v > V).

In the current system, terminal movement is recast as feature interpretation steps, a process unavailable in intermediate movement steps. Therefore, I reformulate the GBOIM as a constraint specifically imposing on possible sequences of feature interpretation steps, while ignoring intermediate movement steps, as shown in (52).

(52) Generalized Ban on Improper Feature Interpretation (GBOIFI)
No constituent may undergo movement for the interpretation of feature $[\sigma]$ if it has undergone movement for the interpretation of feature $[\tau]$, where $[\sigma] \ll [\tau]$ under UCOOL.

In the following, I will show that, under the instantiation of UCOOL in Mandarin Chinese, this weaker version correctly predicts the (im)possible movement paths in the *lian...dou* construction.

4 The instantiation of UCOOL in Mandarin Chinese

In Mandarin Chinese, features and their associated operations are hierarchically ordered as in (53). The crucial point for the present analysis is that features within the composite [D/Foc] probe rank lower than [Topic], which I will argue to be the triggers for TP-internal and TP-external movement, respectively. The remaining features play no role in the analysis; they are included to indicate that Mandarin Chinese may exhibit a total ordering of features.

(53) The Constraint on the Ordering of Operations in Mandarin Chinese
$[\phi] \ll [\mathrm{D}] \ll [\mathrm{Evidentiality}] \ll [\mathrm{Foc}] \ll [\mathrm{Topic}]$

The data is summarized in Table 1, with feeding operations listed in the first column and subsequent movements in the first row (for comparable patterns of feeding relations in different languages, see Williams 2003; Abels 2007; Neeleman and van de Koot 2010). Each cell in the row specifies whether that movement feeds the subsequent movement corresponding to the column of that cell. A check ✓ indicates that an operation feeds the subsequent movement in the corresponding column. An asterisk * indicates that an operation cannot feed the subsequent movement in that column. The notation $\diagdown$ marks where the facts are indeterminate or the interactions are ruled out for independent reasons. For example, the second row of Table 1 states that the

Table 1 Feeding relations between movement operations in Mandarin Chinese

This ↓ feeds this →	[ϕ]	[D]	[Evidentiality]	[Foc]	[Topic]
[ϕ]	—	✓	✓	—	✓
[D]	*	—	✓	—	✓
[Evidentiality]	*	*	—	—	✓
[Foc]	*	*	*	—	✓
[Topic]	*	*	*	*	—

operation involving [ϕ] valuation feeds movement operations triggered by [D], [Evidentiality], and [Topic]. Topicalization in the last row represents the extreme case on the other end. It feeds none of the other four types of movement operations.[16]

In the *bei*-passive construction in Mandarin Chinese, the internal argument is displaced from its base position to the subject position above *bei*, with the external argument optionally surfacing below *bei*, as in (54). Under the analysis proposed by Huang et al. (2009) (see also Ting 1995, 1998; Huang 1999; Tang 2001; Feng 2012; Bruening and Tran 2015), the head *bei* introduces a base-generated subject in its specifier and selects a verbal projection involving movement of a null operator (NOP), as depicted in (55). The subject of *bei* is interpreted as the internal argument of the verb via coindexation with the moved NOP, on a par with Chomsky's (1977) analysis of English *tough*-constructions.[17] To implement this analysis in the feature-based framework, I assume that the introduction of the subject is licensed by the valuation of [ϕ] between *bei* and its specifier.[18]

(54) Lisi$_1$ bei (jingcha) zhua-zou le t$_1$.
Lisi PASS police arrest-away ASP
'Lisi was arrested (by the police).'

(55) $\text{DP}_1\ \text{BEI}_{[\phi]}\ [_{\text{vP}}\ \text{NOP}_1\ \text{DP} \ldots \text{V} \ldots \text{t}_1]$

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[16]The careful reader may note that, in the fifth column, the first three features do not feed subsequent movement triggered by [Foc]. These features do not interact with [Foc] for independent reasons. As shown by Cao (2025a), features within the composite [D/Foc] probe cannot be split: they cannot independently trigger movement and receive interpretation in distinct positions. Consequently, the movement chain in (i)—where movement triggered by [D] is extended by subsequent movement of [Foc]—is impossible. For [Evidentiality], if it feeds movement involving [Foc], the resulting movement chain in (ii) is improper because [D], being bundled with [Foc], is interpreted after the interpretation of [Evidentiality]. Finally, [Foc] triggers movement of VP foci. Since VP foci are not goals for the [ϕ] probe, they remain non-interactive.

(i) $^*[_{\text{XP}}\ \text{lian-DP}_{[\text{D, Foc}]}\ [_{\text{X}'}\ \text{X}_{[\text{Foc}]} \ldots [_{\text{YP}}\ \text{t}'_{[\text{Foc}]}\ [_{\text{Y}'}\ \text{Y}_{[\text{D}]} \ldots$

(ii) $^*[_{\text{XP}}\ \text{lian-DP}_{[\text{D, Foc}]}\ [_{\text{X}'}\ \text{X}_{[\text{D/Foc}]} \ldots [_{\text{YP}}\ \text{t}'_{[\text{D/Foc}]}\ [_{\text{Y}'}\ \text{Y}_{[\text{Evidentiality}]} \ldots$

Nevertheless, this absence of positive evidence does not undermine the ordering in (53). It is widely accepted that [A] probes consistently rank lower than [Ā] probes. Given that the first three probes belong to the A-system (as I will show below), while [Foc] is an [Ā]-feature, the fact that [Foc] ranks higher than the three [A] probes should not be disputed.

[17]Huang et al. (2009) propose that when the external argument is absent, the complement of *bei* involves movement of a PRO, controlled by the base-generated subject. Since we are concerned with operations targeting the subject of *bei*, the derivation within its complement is orthogonal to the present discussion.

[18]The reader should note that [ϕ] is merely a placeholder for whatever feature is associated with the *bei*-passive. The interactions between the *bei*-passive and other movement operations discussed below do not hinge on this choice (for overviews of alternative proposals, see Chen 2023; Ngui 2024).

Bei-passivization feeds movement triggered by the [D/Foc] probe, but not the reverse. In (56a), the passivized DP can undergo subsequent movement to the specifier of *dou*. In contrast, in (56b), the *lian*-DP cannot undergo *bei*-passivization.[19]

(56) a. Lian Lisi$_1$ **dou** t$'_1$ **bei** (jingcha) zhua-zou le t$_1$.
lian Lisi dou PASS police arrest-away ASP
'Even Lisi was arrested (by the police).'

b. * Lian Lisi$_1$ **bei** t$'_1$ **dou** (jingcha) zhua-zou le t$_1$.
lian Lisi dou PASS police arrest-away ASP
'Even Lisi was arrested (by the police).'

The [Evidentiality] feature is responsible for optional raising in constructions where the matrix verb encodes indirect evidentiality, such as *ju-shuo* 'accord-say' in example (57) (For evidence in support of a movement analysis, see Lee and Yip 2024). One evidence for A-properties of [Evidentiality] is that the raised DP can bind an anaphor at the landing site, as shown in (58). The contrast between (59a) and (59b) shows that *bei*-passive feeds raising triggered by [Evidentiality], but not vice versa.

(57) a. Ju-shuo Lisi hen taoyan Zhangsan.
accord-say Lisi very hate Zhangsan
'It is said that Lisi hates Zhangsan deeply.'

b. Lisi$_1$ ju-shuo t$_1$ hen taoyan Zhangsan.
Lisi accord-say very hate Zhangsan
'It is said that Lisi hates Zhangsan deeply.'

(58) Zhangsan$_1$ ju ta-ziji$_1$ de pengyou shuo t$_1$ ma guo Lisi.
Zhangsan accord s/he-self POSS frined say scold EXP Lisi
'As for Zhangsan$_1$, according to his$_1$ own friends, he$_1$ scolded Lisi.'

(59) a. Zhansgan$_1$ ju-shuo t$_1$ **bei** kaichu le.
Zhangsan accord-say PASS fire ASP
"It is said that Zhangsan was fired.'

b. * Zhansgan$_1$ **bei** t$'_1$ ju-shuo t$_1$ kaichu le.
Zhangsan PASS accord-say fire ASP
'It is said that Zhangsan was fired.'

A strong argument that *bei*-passive asymmetrically feeds topicalization is as follows. Examples in (60) show that *bei*-passive creates new antecedents for anaphor binding. In (61), while direct topicalization does not create new antecedents for anaphor binding (61a), topicalization of the subject in a *bei*-passive sentence does

[19] One may take the contrast between (56a) and (56b) as evidence for a cartographic template in Mandarin Chinese, where the projection of *bei* is generated lower than that of *dou* (Rizzi 1997, 2004b). However, an account along this line is untenable. A cartographic template, if it exists, constrains not only the movement path of an individual item, but also the relative positions of the focus and the passivized argument when the two do not coincide. However, as shown in (i), the head *dou* may surface below the head *bei*:

(i) Zhangsan **bei** Lihua ba jiehun jiezhi **dou** pian-zou le.
Zhangsan PASS Lihua BA wedding ring dou scam-away ASP
'As for Zhangsan, even his wedding ring was swindled by Lihua.'

(61b). This contrast is straightforwardly explained if topicalization, while unable to create new antecedents on its own, can be fed by *bei*-passivization.

(60) a. * (ta)-ziji$_1$ de pengyou ma guo Zhangsan$_1$.
(s/he)-self POSS frined scold EXP Zhangsan
Intended: 'Zhangsan$_1$, his$_1$ own friends scolded him$_1$.'

b. Zhangsan$_1$ bei (ta)-ziji$_1$ de pengyou ma guo t$_1$.
Zhansgan PASS (s/he)-self friend POSS scold EXP
'Zhangsan$_1$ was scolded by his$_1$ own friends.'

(61) a. * Zhangsan$_1$-a [$_{\text{TP}}$ (ta)-ziji$_1$ de pengyou ma guo t$_1$].
Zhangsan-TOP (s/he)-self POSS frined scold EXP
Intended: 'As for Zhangsan$_1$, his$_1$ own friends scolded him$_1$.'

b. Zhangsan$_1$-a [$_{\text{TP}}$ t$'_1$ bei (ta)-ziji$_1$ de pengyou ma guo t$_1$].
Zhansgan-TOP PASS (s/he)-self friend POSS scold EXP
'As for Zhangsan$_1$, he$_1$ was scolded by his$_1$ own friends.'

In example (62a), the *lian*-DP undergoes raising triggered by [Evidentiality]. In contrast, as highlighted by the position of *dou* in (62b), the raised DP cannot undergo subsequent movement triggered by [D/Foc].

(62) a. Lian Zhangsan$_1$ ju-shuo t$_1$ **dou** jige le.
lian Zhangsan accord-say dou pass ASP
'It is said that even Zhangsan passed the exam.'

b. * Lian Zhangsan$_1$ **dou** t$'_1$ ju-shuo t$_1$ jige le.
lian Zhangsan dou accord-say pass ASP
'It is said that even Zhangsan passed the exam.'

As shown in (63), the TP-internal *lian*-DP cannot take the topic *a* (63a), whereas the TP-external *lian*-DP can (63b). This contrast suggests that the TP-external *lian*-DP is derived by topicalization targeting the TP-internal *lian*-DP (Shyu 1995; Badan 2008; Badan and Del Gobbo 2015).[20]

(63) a. * Zhangsan lian [Li jiaoshou-a]$_1$ dou bu xinren t$_1$.
Zhangsan lian Li professor-TOP dou NEG trust
'Zhangsan does not trust even Professor Li.'

b. Lian [Li jiaoshou-a]$_1$ Zhangsan t$'_1$ dou bu xinren t$_1$.
lian Li professor-TOP Zhangsan dou NEG trust
'Even Professor Li, Zhangsan does not trust (him/her).'

Additional support comes from reconstruction effects we saw in Sect. 2. Example (64) shows that TP-internal movement of DP foci does not reconstruct. In contrast,

[20] In English, foci associated with *even* cannot be topicalized via the "as for" construction, as shown in (i). I argue that example (i) is unacceptable because the particle *even* fails to take propositional scope. First, the particle *even* cannot take scope over the comment clause because it is embedded within the "as for" phrase. Second, "as for" introduces a base-generated topic. *Even* cannot reconstruct to a scope position within the comment clause. In contrast, as we will see below, TP-external *lian*-phrases reconstruct to the position where the focus is interpreted.

(i) * As for even Professor Li, Zhangsan does not trust (him/her).

topicalization reconstructs for anaphor binding, as in (65). In (66), the anaphor within the TP-external *lian*-DP can only be bound by the matrix subject *Zhang*, not by the lower argument *Li*. This limited reconstruction effect suggests that movement triggered by [D/Foc] can be fed by topicalization, with the latter licensing reconstruction.

(64) * Wo lian [ta-ziji$_1$ qian de qian]$_2$ dou ti Lihua$_1$ huanqing le t$_2$.
I lian s/he-self owe REL money dou for Lihua pay ASP
Intended: 'I helped Lihua$_1$ pay off even the debt he$_1$ owed himself$_1$.'

(65) [Ta-ziji$_1$ de pengyou]$_2$(-a) Zhangsan$_1$ changchang piping t$_2$.
s/he-self POSS friend TOP Zhangsan often criticize
'As for his$_1$ own friend, Zhangsan$_1$ always criticizes them.'

(66) Lian [$_{DP}$ ta-ziji$_{1/*2}$ de pengyou]$_3$(-a) Zhang$_1$ t$'_3$ dou bi Li$_2$ [$_{TP}$ ma guo t$'_3$].
lian s/he-self POSS friend TOP Zhang dou force Li scold EXP
'Even his$_{1/*2}$ own friends, Zhang$_1$ forced Li$_2$ to scold (them).'

The interaction between [Foc] and [Topic] is illustrated with sub-extraction out of a moved category. In example (67), topicalization targets the object within the TP-internal *lian*-VP. In contrast, in example (69), sub-extraction triggered by [Foc] cannot target the topicalized VP within the embedded CP. Given that the VP can be topicalized within its embedded clause (68), the contrast between (67) and (69) provides strong evidence that [Foc] asymmetrically feeds [Topic].

(67) [$_{DP}$ Zhe-ben shu-a]$_1$ Zhang lian [$_{VP}$ wanzheng-de kan yi-bian t$_1$]$_2$ dou bu yuanyi t$_2$.
This-CLF book-TOP Zhang lian completely-ADV read one-time dou NEG willing
'As for this book, Zhang is not willing to read it even once.'

(68) Li juede [$_{CP}$ [$_{VP}$ bi Yu [$_{VP}$ wanzheng-de kan yi-bian zhe-ben shu(-ne)]]$_1$ Wu juedui bu yuanyi t$_1$].
Li think force Yu completely-ADV read one-time this-CLF book-TOP Wu definitely NEG willing
'Li thinks that force Yu to read this book once, Wu is definitely not willing to.'

(69) * Lian [$_{VP}$ wanzheng-de kan yi-bian zhe-ben shu]$_2$ Li dou juede [$_{CP}$ [$_{VP}$ bi Yu zuo(-ne) t$_2$]$_1$ Wu juedui bu yuanyi t$_1$].
Lian completely-ADV read one-time this-CLF book Li dou think force Yu do-TOP Wu definitely NEG willing
Intended: "Li thinks that force Yu to even read this book once, Wu is definitely not willing to.'

5 Application of the proposal

5.1 TP-internal movement in the *lian... dou* construction

(70) (Im)proper cross-clausal movement in the *lian... dou* construction

a. lian-XP$_1$ Subject t'_1 dou ... [$_{TP}$ ... t_1 ...

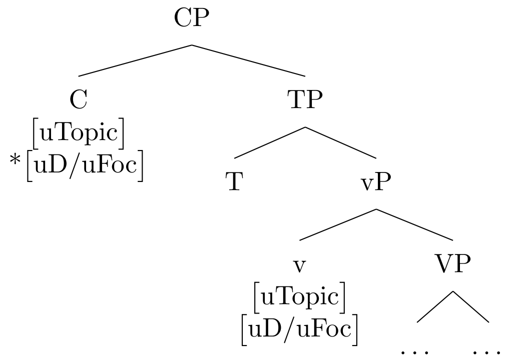
b. lian-XP$_1$ Subject t'_1 dou ... [$_{CP}$ ... [$_{TP}$ ... t_1 ...

c. Subject lian-XP$_1$ dou ... [$_{TP}$ ... t_1 ...

d. * Subject lian-XP$_1$ dou ... [$_{CP}$ ... [$_{TP}$ ... t_1 ...

The pattern of (im)proper cross-clausal movement in the *lian... dou* construction is repeated in (70). TP-external movement is triggered by [Topic], which is capable of crossing TPs (70a) and CPs (70b). In contrast, TP-internal movement involves a [uD/uFoc] probe. It is more restricted, proceeding only out of TPs (70c-d). I attribute this locality mismatch to the different distributions of [uD/uFoc] and [uTopic] probes in Mandarin Chinese. As outlined in (71), while [uTopic] probes are borne by both C and v heads, [uD/uFoc] probes are present only on v heads.[21]

(71)



(73) illustrates the derivation of legitimate TP-internal movement, such as example (72). Movement proceeds cyclically and terminates at the matrix [Spec, vP], where [D/Foc] is interpreted.

(72) Li lian [$_{DP}$ zhe-ben shu]$_1$ dou bi [$_{TP}$ Wu kan-wan le t_1].
Li lian this-CLF book dou force Wu read-finish ASP
'Li forced Wu to read even this book.'

(73) a. [$_{v'}$ $v_{[uD/uFoc]}$ [... lian-DP$_{[D,\ Foc]}$ ...

b. [$_{vP}$ lian-DP$_{[D,\ Foc]}$ [$_{v'}$ $v_{[D/Foc]}$ ...

c. [$_{TP}$ T [$_{vP}$ lian-DP$_{[D,\ Foc]}$ [$_{v'}$ $v_{[D/Foc]}$ ...

d. [$_{v'}$ $v_{[uD/uFoc]}$ [$_{TP}$ T [$_{vP}$ lian-DP$_{[D,\ Foc]}$ [$_{v'}$ $v_{[D/Foc]}$ ...

e. [$_{vP}$ lian-DP$_{[D,\ Foc]}$ [$_{v'}$ $v_{[D/Foc]}$ [$_{TP}$ T [$_{vP}$ $t'_{[D/Foc]}$ [$_{v'}$ $v_{[D/Foc]}$ ...

[21] Given that the discussion here concerns successive-cyclic movement through phase edges, I assume, for simplicity, that v heads are overtly realized as *dou* when they bear [uD/uFoc] probes. The core argument is compatible with the analyses of the *lian... dou* construction by Chen (2023) and Cao (2025a), in which *dou* is a TP-internal focus projection higher than vPs, both of which host [uD/uFoc] probes.

f. [$_{\text{TP}}$ T [$_{\text{vP}}$ lian-DP$_{\text{[D, Foc]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc]}}$ [$_{\text{TP}}$ T [$_{\text{vP}}$ t$'_{\text{[D/Foc]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc]}}$ ...

TP-internal movement cannot cross CPs because C heads do not bear [uD/uFoc] probes. (75) shows a possible derivation of (74) where the *lian*-DP moves directly to the higher [Spec, vP] without stopping at the embedded [Spec, CP]. This is ruled out by the PIC. When the higher v is merged, the *lian*-DP is trapped in the domain of the embedded CP and becomes inaccessible to the [uD/uFoc] probe on the higher v head. (76) presents an alternative derivation in which the *lian*-DP moves through the embedded [Spec, CP]. Although this derivation obeys the PIC, movement to the embedded [Spec, CP] violates the Last Resort condition.

(74) * Lisi lian laobao$_1$ dou juede [$_{\text{CP}}$ Zhangsan ma guo t$_1$]
Lisi lian boss dou think Zhangsan scold EXP
'Intended: 'Even the boss$_1$, Lisi thinks Zhangsan scolded him$_1$.'

(75) Violations of the Phase Impenetrability Condition
*[$_{\text{vP}}$ lian-DP$_{\text{[D, Foc]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc]}}$ [$_{\text{CP}}$ C [$_{\text{vP}}$ t$'_{\text{[D/Foc]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc]}}$ ...

(76) Violation of Last Resort at CP
*[$_{\text{vP}}$ lian-DP$_{\text{[D, Foc]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc]}}$ [$_{\text{CP}}$ t$''_{\text{[D/Foc]}}$ [$_{\text{C}'}$ C [$_{\text{vP}}$ t$'_{\text{[D/Foc]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc]}}$ ...

5.2 TP-external movement in the *lian... dou* construction

Example (77) shows TP-external movement out of a TP. The derivation begins with terminal movement driven by [D/Foc], extended by movement triggered by [Topic]. In (78a) to (78e), cyclic movement is driven by [D/Foc] and [Topic] is shared as a free rider. [D/Foc] is interpreted at the matrix vP (78e). Subsequent movement is driven by [Topic], as in (78g). Since the order in which [D/Foc] and [Topic] are interpreted conforms to the GBOIFI, the derivation in (78) is well-formed.

(77) Lian [$_{\text{DP}}$ zhe-ben shu]$_1$ Li dou bi [$_{\text{TP}}$ Wu kan-wan le t$_1$].
lian this-CLF book Li dou force Wu read-finish ASP
'Li forced Wu to read even this book.'

(78) a. [$_{\text{v}'}$ v$_{\text{[uD/uFoc], [uTopic]}}$ [... lian-DP$_{\text{[D, Foc, Topic]}}$ ...
b. [$_{\text{vP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ ...
c. [$_{\text{TP}}$ T [$_{\text{vP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ ...
d. [$_{\text{v}'}$ v$_{\text{[uD/uFoc], [uTopic]}}$ [$_{\text{TP}}$ T [$_{\text{vP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ ...
e. [$_{\text{vP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ [$_{\text{TP}}$ T [$_{\text{vP}}$ t$'_{\text{[D/Foc], [Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ ...
f. [$_{\text{C}'}$ C$_{\text{[uTopic]}}$ [$_{\text{vP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [uTopic]}}$ [$_{\text{TP}}$ T [$_{\text{vP}}$ t$'_{\text{[D/Foc], [Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ ...
g. [$_{\text{CP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ [$_{\text{C}'}$ C$_{\text{[Topic]}}$ [$_{\text{vP}}$ t$'_{\text{[Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [uTopic]}}$ [$_{\text{TP}}$ T [$_{\text{vP}}$ t$'_{\text{[D/Foc], [Topic]}}$ [$_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ ...

Let us now turn to long-distance TP-external movement, such as example (79). A puzzle we have seen previously arises from the contrast between long-distance TP-external movement in (80a) and long-distance TP-internal movement in (80b). On the one hand, movement triggered by [D/Foc] cannot proceed out of CPs on its own (80b). This rules out an analysis in which C heads bear [uD/uFoc] probes. If the system were to allow this, it would overgenerate ungrammatical cases such as those schematized in (81), in which the *lian*-DP moves through the embedded [Spec, CP] and surfaces at the matrix [Spec, vP]. On the other hand, it is not the case that [D/Foc] cannot be interpreted in the matrix clause. In (80a), the *lian*-phrase moves out of the embedded CP and values the [uD/uFoc] probe at a TP-internal position t'_1 within the matrix clause. This peculiar movement path suggests that movement feeding the interpretation of [D/Foc] may cross CP boundaries only if it is extended by topicalization.

(79) Lian laobao$_1$ Lisi t'_1 dou juede [$_{CP}$ Zhangsan ma guo t_1]
lian boss Lisi dou think Zhangsan scold EXP
'Even the boss$_1$, Lisi thinks Zhangsan scolded him$_1$.'

(80) a. lian-XP$_1$ Subject t'_1 dou ... [$_{CP}$ ... [$_{TP}$ ... t_1 ...
b. * Subject lian-XP$_1$ dou ... [$_{CP}$ ... [$_{TP}$ ... t_1 ...

(81) * [$_{TP}$ T [$_{vP}$ lian-DP$_{[D,\ Foc]}$ [$_{v'}$ v$_{[D/Foc]}$ [$_{CP}$ $t''_{[D/Foc]}$ [$_{C'}$ C$_{[D/Foc]}$ [$_{vP}$ $t'_{[D/Foc]}$ [$_{v'}$ v$_{[D/Foc]}$ ...

I propose that, in examples like (79), movement to the embedded [Spec, CP] is triggered by [Topic], and the subsequent [Spec, CP]-to-[Spec, vP] movement is triggered by [D/Foc]. This derivation may initially appear counterintuitive, as movement triggered by [Topic] is followed by movement triggered by the lower-ranked feature [D/Foc]. However, as I will show below, this derivation is well-formed under the GBOIFI. Specifically, (82) illustrates the derivation within the embedded clause. The *lian*-DP moves cyclically to the embedded [Spec, CP], with each step driven by [Topic].

(82) a. [$_{v'}$ v$_{[uTopic]}$ [... lian-DP$_{[D,\ Foc,\ Topic]}$ ...
b. [$_{vP}$ lian-DP$_{[D,\ Foc,\ Topic]}$ [$_{v'}$ v$_{[Topic]}$ ...
c. [$_{C'}$ C$_{[uTopic]}$ [$_{vP}$ lian-DP$_{[D,\ Foc,\ Topic]}$ [$_{v'}$ v$_{[Topic]}$ ...
d. [$_{CP}$ lian-DP$_{[D,\ Foc,\ Topic]}$ [$_{C'}$ C$_{[Topic]}$ [$_{vP}$ $t'_{[Topic]}$ [$_{v'}$ v$_{[Topic]}$ ...

In (83b), movement from the embedded [Spec, CP] to the matrix [Spec, vP] results in the valuation of [D/Foc] and [Topic]. This is the interpretation step for [D/Foc] because the higher C head, which provides a landing site for subsequent movement, does not bear a [uD/uFoc] probe. As a result, [D/Foc] is interpreted on the matrix v head. However, the GBOIFI is not invoked at this stage, as the embedded [Spec, CP] is an intermediate step for [Topic]. It is only when all relevant features have been interpreted that the GBOIFI becomes operative. As shown in (83d), When the *lian*-DP reaches the matrix [Spec, CP], [Topic] is interpreted there. Since the movement step interpreting [D/Foc] is followed by the movement step interpreting [Topic], this movement path is permitted by the GBOIFI.

(83) a. [$_{v'}$ v$_{[uD/\text{UEVEN}],\ [uTopic]}$ [$_{CP}$ lian-DP$_{[D,\ Foc,\ Topic]}$ [$_{C'}$ C$_{[Topic]}$ [$_{vP}$ $t'_{[Topic]}$ [$_{v'}$ v$_{[Topic]}$ ...

b. $[_{\text{vP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ $[_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ $[_{\text{CP}}$ t$''_{\text{[Topic]}}$ $[_{\text{C}'}$ C$_{\text{[Topic]}}$ $[_{\text{vP}}$ t$'_{\text{[Topic]}}$ $[_{\text{v}'}$ v$_{\text{[Topic]}}$ $\ldots$

c. $[_{\text{C}'}$ C$_{\text{[uTopic]}}$ $[_{\text{vP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ $[_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ $[_{\text{CP}}$ t$''_{\text{[Topic]}}$ $[_{\text{C}'}$ C$_{\text{[Topic]}}$ $[_{\text{vP}}$ t$'_{\text{[Topic]}}$ $[_{\text{v}'}$ v$_{\text{[Topic]}}$ $\ldots$

d. $[_{\text{CP}}$ lian-DP$_{\text{[D, Foc, Topic]}}$ $[_{\text{C}'}$ C$_{\text{[Topic]}}$ $[_{\text{vP}}$ t$'''_{\text{[Topic]}}$ $[_{\text{v}'}$ v$_{\text{[D/Foc], [Topic]}}$ $[_{\text{CP}}$ t$''_{\text{[Topic]}}$ $[_{\text{C}'}$ C$_{\text{[Topic]}}$ $[_{\text{vP}}$ t$'_{\text{[Topic]}}$ $[_{\text{v}'}$ v$_{\text{[Topic]}}$ $\ldots$

6 Cyclic movement and clause-internal operations

We have seen that, in Mandarin Chinese, the [D/Foc] probe cannot establish syntactic dependencies across CP boundaries, whereas topicalization can. However, this does not mean that the [D/Foc] probe never interacts with an embedded goal. TP-external movement in the *lian... dou* construction instantiates a particular type of movement path in which successive-cyclic movement driven by topicalization feeds a matrix scope interpretation of focus at an intermediate landing site. Crucially, such movement paths are possible only under the GBOIFI, which treats intermediate positions as accessible for independent syntactic operations.

Although not all languages exhibit this kind of movement path, it is attested in the languages listed in Table 2.[22] These languages display the following characteristics. The movement operations in the third column, can target constituents embedded within CPs. By contrast, the local operations in the final column cannot establish syntactic dependencies across CP boundaries. However, an embedded constituent may nonetheless be targeted by the local operation, despite normally being inaccessible, once the constituent undergoes long-distance movement. In other words, in these languages, the local operation can be parasitic on successive-cyclic movement because intermediate movement steps place the moving constituent into the local configuration that licenses the operation. Notably, neither the movement operation nor the local operation is restricted to any single type. This suggests that these patterns are not attributable to the specific movement trigger; rather, they reflect general properties of long-distance movement. Namely, long-distance movement must proceed cyclically through clause-internal intermediate landing sites, where an independent local operation can apply. To illustrate the diversity of movement and local operations, consider the Koryak data.

In Koryak, the calculus of case follows the algorithm in (84) (a version tailored for the Koryak data; for a more general formulation of case calculus in dependent case theory, see Poole 2023 and references therein).

(84) The calculus of case in Koryak

a. Assign idiosyncratic lexical and inherent cases.

b. Take the remaining DPs. If DP$_\alpha$ c-commands DP$_\beta$ in the same local domain, assign ergative case to DP$_\alpha$.

[22] Data from languages other than Mandarin are drawn from the literature: Hungarian (den Dikken 2009, pear; Gervain 2009; Jánosi 2014; Jánosi et al. 2014), Koryak (Abramovitz 2020), Kiribati (Sabel 2013), Standard Indonesian Sato (2012) and Asante Twi (Korsah and Murphy 2020).

c. If a DP was not assigned a case in the previous two steps, then assign it unmarked case (absolutive).

Table 2 Interactions between movement and local operations

Language	domain	Movement operation	Local operation
Mandarin Chinese	CP	Topicalization	Focus interpretation
Hungarian	CP	Focus/*wh*-mvt	Case assignment
Koryak	CP	*Wh*-mvt	Case assignment
Kiribati	CP	Focus mvt	Object agreement on verbs
Standard Indonesian	CP	*Wh*-mvt, relativization	Active voice morphology
Asante Twi	CP	Focus mvt, relativization	High tone overwriting on verbs

For readability, I omit the glossing of the verbal morphology in the Koryak data. In example (85), the transitive verb does not assign its complement a lexical case. Under the second rule in (84b), the subject is marked ergative and the object absolutive. By contrast, in (86), the verb assigns its object the narrative case, a lexical case in Koryak. The subject then receives the unmarked absolutive case.

(85) (Abramovitz 2020, p. 7)
ɣəmnan tənunew ʔəvənʔ-u
1SG.ERG eat berry-ABS.PL
'I ate berries.'

(86) (Abramovitz 2020, p. 8)
ɣəmmo təkemŋoləØ kəmiŋə-kjit.
1SG.ABS miss child-NARR
'I miss my son.'

Example (87) illustrates the case frame of verbs that take a finite clausal complement. The embedded verb does not assign lexical case, yielding the same case frame in the embedded clause as in the basic transitive clause in (85). The matrix subject receives the absolutive case because it is the only DP in the matrix clause.

(87) (Abramovitz 2020, p. 17)
ɣəmmo təvalomək əno ʔewngəto-na-k Øjətçimawnin kojŋ-o
1SG.ABS hear that Hewngyto-OBL.SG-ERG break cup-ABL.PL
'I heard that Hewngyto broke cups.'

Notably, if the embedded (absolutive) object undergoes *wh*-movement to the left periphery, as in (88), the matrix subject must instead bear ergative case.

(88) (Abramovitz 2020, p. 17)
jej-u$_1$ {ɣə-nan / *ɣətçtç} Øvalomnaw əno ʔewŋəto-na-k
what-ABS.PL {2SG-ERG / *2SG.ABS} hear that Hewngyto-OBL.SG-ERG
Øjətçimawnin t$_1$
break
'What all did you hear that Hewngyto broke?'

In principle, the moving *wh*-phrase may trigger ergative case on the otherwise-absolutive subject either when it stops at the edge of the embedded CP or when it reaches a higher intermediate landing site within the matrix clause. The sentences in example (89) suggest that only the latter position licenses ergative case. In each sentence in (89), the embedded clause is an indirect question in which an absolutive *wh*-phrase moves to the embedded [Spec, CP]. However, the matrix subject must bear absolutive case (89a), rather than ergative case (89b). This contrast indicates that in Koryak, the assignment of ergative case happens at an intermediate landing site within the matrix clause, where the moving DP and the matrix subject stand in a c-command relation.

(89) (Abramovitz 2020, p. 30)

a. ɣəmmo təkutçetkejuŋəŋ∅ [$_{\text{CP}}$ jeq-qevi-jətçɣ-u məjəlnew
1SG.ABS think which-gift-contents-ABS.PL give
ʔewŋəto-na-ŋ ənək-eto-ɣəlwəje-ŋ.]
Hewngyto-OBL.SG-DAT 3SG.POSS-birth-day-DAT
'I am wondering what gifts I should give Hewngyto for his birthday.'

b. * ɣəm-nan təkutçetkejuŋŋənew [$_{\text{CP}}$ jeq-qevi-jətçɣ-u məjəlnew
1SG-ERG think which-gift-contents-ABS.PL give
ʔewŋəto-na-ŋ ənək-eto-ɣəlwəje-ŋ.]
Hewngyto-OBL.SG-DAT 3SG.POSS-birth-day-DAT
'I am wondering what gifts I should give Hewngyto for his birthday.'

By contrast, such movement paths are unavailable in English. Example (90) shows that *wh*-movement cannot license ϕ-agreement on matrix T. This is because, in English, the matrix T is not an intermediate landing site of long-distance *wh*-movement. As a result, long-distance *wh*-movement cannot form a continuous chain that passes through matrix T. (91) illustrates the derivation in which the *wh*-phrase moves from the matrix [Spec, vP] to the matrix [Spec, TP]. Since matrix T does not bear an [uWh] probe, [Wh] is interpreted on the matrix v. After *who* moves to the matrix [Spec, TP], [ϕ] on the matrix T is interpreted. Under GBOIFI, this movement path is illicit because [ϕ] cannot be interpreted after [Wh] has been interpreted. Alternatively, since TP is not a phase, *wh*-movement can proceed directly from the matrix [Spec, vP] to the matrix [Spec, CP], as in (92). In this case, the [ϕ] probe on T remains unvalued, resulting in an illicit object at the interface.

(90) * Who$_1$ does t$'_1$ seem [$_{\text{CP}}$ t$_1$ has left]?

(91) Violations of the GBOIFI
*[$_{\text{TP}}$ Who$_{[\phi],[\text{Wh}]}$ [$_{\text{T}'}$ T$_{[\phi]}$ [$_{\text{vP}}$ t$'_{[\phi]}$ [$_{\text{v}'}$ v$_{[\text{Wh}]}$ [$_{\text{CP}}$ t$'_{[\text{Wh}]}$ [$_{\text{C}'}$ C$_{[\text{Wh}]}$ ...

(92) Unvalued features at the interface
*[$_{\text{CP}}$ Who$_{[\phi],[\text{Wh}]}$ [$_{\text{C}'}$ C$_{[\text{Wh}]}$ [$_{\text{TP}}$ T$_{[\text{u}\phi]}$ [$_{\text{vP}}$ t$'_{[\text{Wh}]}$ [$_{\text{v}'}$ v$_{[\text{Wh}]}$ [$_{\text{CP}}$ t$'_{[\text{Wh}]}$ [$_{\text{C}'}$ C$_{[\text{Wh}]}$ ...

7 Deriving the Height-Locality connection

Previous studies on movement dependencies have unearthed an intriguing generalization concerning locality mismatches across movement types. Namely, there appears to be a connection between the locality of a movement type (short vs. long) and the structural height of that movement type's landing site (low vs. high), as stated in (93). (see e.g., Williams 2003, 2013; Müller and Sternefeld 1993; Abels 2007, 2012a,b; Neeleman and van de Koot 2010; Müller 2014a,b; Keine 2016, 2019, 2020; Meadows 2024).[23]

(93) Height-Locality Connection (HLC) (adapted from Keine 2020, p. 12)
The higher the landing site of a movement type is in the *fseq*; the more kinds of structures over which this movement type can apply.

The WC captures the HCL by arguing that the height of a movement's final landing site fully determines its locality property. It predicts that movement of the general shape in (94) is not possible. That is, movement that targets YP cannot move through or cross XP if XP is higher than YP in the *fseq*.

(94) $[_{\text{XP}}$ X … $[_{\text{YP}}$ α $[_{\text{Y}'}$ Y … $[_{\text{XP}}$ X … $[_{\text{YP}}$ t_α $[_{\text{Y}'}$ Y …

(95) * $[_{\text{TP}}$ John$_1$ seems $[_{\text{CP}}$ t_1 has left$]]$

Although the WC correctly rules out illegitimate raising in English, as in (95), movement constructions that instantiate the configuration in (96) are indeed attested (see Williams 2003; Abels 2007, 2012a; Müller 2014a; Keine 2020). One particular case is hyperraising (see Zyman 2023 for a recent overview). For example, in the Bantu language Lubukusu, A-movement can cross CP boundaries. In (96a), the embedded verb bears the same noun class marker '2' as the embedded subject *babaandu*, but the matrix verb is marked with '6'. This indicates that an in-situ subject agrees only with embedded T. In contrast, in (96b), both of the verbs are assigned the marker '2', due to movement of *babaandu*. That the moved subject controls matrix verb agreement indicates that it is A-movement.

(96) (Carstens and Diercks 2013, p. 100)

a. $[_{\text{TP}}$ Ka-lolekhana $[_{\text{CP}}$ mbo babaandu ba-kwa $]]$
6SA-seem that 2people 2SA.PST-fall
'It seems that the people fell.'

b. $[_{\text{TP}}$ babaandu$_1$ ba-lolekhana $[_{\text{CP}}$ mbo t_1 ba-kwa $]]$
2people 2SA-seem that 2SA.PST-fall
'The people seem like they fell.'

A similar cross-linguistic variation arises in movement targeting the middle field of the clause, commonly referred to as scrambling. In the *lian… dou* construction, TP-internal movement cannot proceed out of CPs (97). By contrast, in Russian, scrambling can proceed out of CPs, as in (98).

(97) * Lisi lian laobao$_1$ dou juede $[_{\text{CP}}$ Zhangsan ma guo t_1 $]$
Lisi lian boss dou think Zhangsan scold EXP
Intended: 'Even the boss$_1$, Lisi thinks Zhangsan scolded him$_1$.'

[23] The HLC not only holds for the movement path of an individual item, but also covers cases of subextraction and remnant movement (Müller and Sternefeld 1993; Abels 2007, 2012b; Neeleman and van de Koot 2010), as well as syntactic dependencies that do not involve movement, such as long-distance agreement (Keine 2016, 2019, 2020) and case assignment (Poole 2023). In this paper, I focus on the movement path of a single element and leave extensions to related configurations for future work.

(98) (Keine 2020, p. 195)
On [ètu knigu]$_1$ dumajet [$_{\text{CP}}$ čto Pjot pročital t$_1$]
he this book thinks that Pjot read
'He thinks that Pjotr read this book.'

These cross-linguistic data suggest that whether a movement type obeys the HLC should be parameterized. I will not provide an exhaustive derivation of the HLC and all of its apparent exceptions, as this lies well beyond the scope of this paper. Nevertheless, I show that once the locality of a movement type is attributed to the distribution of its triggering features (as implemented throughout the paper), cross-linguistic variation in locality patterns follows as an immediate prediction. In short, languages may differ parametrically in how triggering features are distributed, and locality patterns are predicted to vary accordingly (for alternative approaches, see Obata and Epstein 2011; Müller 2014a; Keine 2019; Poole 2023). In this respect, the feature-based system proposed here contrasts with the WC, which is overly restrictive in that it universally bans movement dependencies that fall outside the HLC.

For expository purposes, the *altitude* of a feature is defined as the highest head in an extended projection that hosts that feature. This notion makes it possible to state an explicit mapping between feature distribution across extended projections. I begin with movement that conforms to the HLC. In such cases, the altitude of the triggering feature is matched across extended projections. To see the consequence of this altitude-matching pattern, suppose that a feature [τ] distributes over C and v.[24] In other words, [τ] has C-level altitude across extended projections, as indicated by the horizontal arrow '$\rightarrow$' in the first line of (99). Under this configuration, movement triggered by [τ] instantiates long-distance movement that targets a high landing site (e.g., *wh*-movement in English). If [τ] is confined to v or T, as in (100–101), movement is clause-bounded and targets a low landing site (e.g., TP-internal movement in the *lian*... *dou* construction and A-movement in English).

(99) The altitude of [τ]: C $\rightarrow$ C
$[_{\text{CP}}\ \alpha_{[\tau]}\ [_{\text{C}'}\ \text{C}_{[\tau]}\ [_{\text{vP}}\ \text{t}'''_{[\tau]}\ [_{\text{v}'}\ \text{v}_{[\tau]}\ [_{\text{CP}}\ \text{t}''_{[\tau]}\ [_{\text{C}'}\ \text{C}_{[\tau]}\ [_{\text{vP}}\ \text{t}'_{[\tau]}\ [_{\text{v}'}\ \text{v}_{[\tau]} \ldots$

(100) The altitude of [τ]: v $\rightarrow$ v
a. $[_{\text{CP}}\ \text{C}\ [_{\text{vP}}\ \alpha_{[\tau]}\ [_{\text{v}'}\ \text{v}_{[\tau]}\ [_{\text{VP}}\ \text{t}_{[\tau]} \ldots$
b. * $[_{\text{CP}}\ \text{C}\ [_{\text{vP}}\ \alpha_{[\tau]}\ [_{\text{v}'}\ \text{v}_{[\tau]}\ [_{\text{CP}}\ \text{C}\ [_{\text{vP}}\ \text{t}'_{[\tau]}\ [_{\text{v}'}\ \text{v}_{[\tau]} \ldots$

(101) The altitude of [τ]: T $\rightarrow$ T
a. $[_{\text{CP}}\ \text{C}\ [_{\text{TP}}\ \alpha_{[\tau]}\ [_{\text{T}'}\ \text{T}_{[\tau]}\ [_{\text{vP}}\ \text{t}_{[\tau]} \ldots$
b. * $[_{\text{CP}}\ \text{C}\ [_{\text{TP}}\ \alpha_{[\tau]}\ [_{\text{T}'}\ \text{T}_{[\tau]}\ [_{\text{CP}}\ \text{C}\ [_{\text{TP}}\ \text{t}'_{[\tau]}\ [_{\text{v}'}\ \text{T}_{[\tau]} \ldots$

Exceptions to the HLC are due to an altitude-mismatching pattern. That is, the altitude of a given feature in one extended projection fails to align with its altitude in other extended projections. In (102) and (103), the altitude of [τ] decreases from C to T and v, respectively, as indicated by the downward arrow '$\searrow$'.[25] Consequently,

[24] The analysis does not depend on the specific identity of [τ]. Rather, we focus on the general shape of movement generated by features of different altitudes.

[25] A third logical possibility is that the altitude of [τ] increases from one extended projection to another: v $\nearrow$ T/C and T $\nearrow$ C. This kind of configuration yields movement targeting higher domains only if the

long-distance movement targets TP and vP domains (e.g., hypperraising in Lubukusu and scrambling in Russian).[26]

(102) The altitude of [τ]: C $\searrow$ T
$[_{\text{CP}}$ C $[_{\text{TP}}$ $\alpha_{[\tau]}$ $[_{\text{T}'}$ $\text{T}_{[\tau]}$ $[_{\text{vP}}$ $\text{t}'''_{[\tau]}$ $[_{\text{v}'}$ $\text{v}_{[\tau]}$ $[_{\text{CP}}$ $\text{t}''_{[\tau]}$ $[_{\text{C}'}$ $\text{C}_{[\tau]}$ $[_{\text{vP}}$ $\text{t}'_{[\tau]}$ $[_{\text{v}'}$ $\text{v}_{[\tau]}$ $\ldots$

(103) The altitude of [τ]: C $\searrow$ v
$[_{\text{CP}}$ C $[_{\text{vP}}$ $\alpha_{[\tau]}$ $[_{\text{v}'}$ $\text{v}_{[\tau]}$ $[_{\text{CP}}$ $\text{t}''_{[\tau]}$ [$\text{C}_{[\tau]}$ $[_{\text{vP}}$ $\text{t}'_{[\tau]}$ $[_{\text{v}'}$ $\text{v}_{[\tau]}$ $\ldots$

Even if the distribution of features is arbitrary, as assumed throughout the paper, the system still constrains the resulting derivations. In particular, features must be distributed such that their interpretation strictly respects the ordering imposed by UCOOL. In other words, although any feature could in principle be assigned any altitude, certain combinations are excluded because they inevitably yield improper movement chains. Suppose that there is another feature [σ], with [σ] $\ll$ [τ] on the UCOOL. Within any extended projection, the altitude of [σ] cannot exceed that of [τ]. (104) illustrates a case in which [σ] has C-level altitude while [τ] has T-level altitude. In the resulting movement chain, the interpretation step for [τ] is followed by the interpretation step for [σ], a sequence that is illicit under GBOIFI.

(104) * $\ldots$ $[_{\text{CP}}$ $\alpha_{[\sigma],\,[\tau]}$ $[_{\text{C}'}$ $\text{C}_{[\sigma]}$ $[_{\text{TP}}$ $\text{t}'_{[\sigma]}$ $[_{\text{T}'}$ $\text{T}_{[\tau]}$ $[_{\text{vP}}$ $\text{t}'_{[\tau]}$ $[_{\text{v}'}$ $\text{v}_{[\tau]}$ $\ldots$

Cross-linguistically, movement conforming to the HLC represents the general pattern. From the perspective of language acquisition, the cross-linguistic robustness of the HLC follows naturally if altitude matching is the default setting. By contrast, altitude mismatching can be acquired only on the basis of positive evidence that departs from the default. Children are born assuming the altitude-matching pattern, in the absence of countervailing evidence. For example, if a feature's altitude is C, the child assumes that it maintains this altitude across extended projections. This yields HLC-obeying movement, such as *wh*-movement in English. Alternatively, the learner may encounter evidence that a feature has C-level altitude in one extended projection but T-level altitude in the next. Based on such evidence, the default altitude-matching setting is overwritten by an altitude-mismatching one. The child then acquires HLC-violating movement, such as hyperraising in Lubukusu.[27]

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category to the left of $\nearrow$ is not c-commanded by a phase head. As far as I can see, this pattern bears on successive cyclicity rather than on improper movement. I therefore leave this aside for reasons of space.

[26] Likewise, the altitude of [τ] may decrease from T to v, as in (i). One may take the English ECM construction as an instance of this configuration, under the analysis in which the embedded subject moves from the embedded TP to a low position in the matrix clause (Postal 1974). This analysis is motivated by the possibility of matrix adverbs intervening between the embedded subject and predicate (e.g., *sincerely* in (ii)). However, on the basis of scope relations and adverb ordering, Neeleman and Payne (2020) argue that ECM infinitives do not involve movement of the embedded subject (ii), but rather rightward partial extraposition of the embedded clause (iii). Since the relevant data instantiating this pattern are largely absent from the literature, I set this issue aside in the main text.

(i) The altitude of [τ]: T $\searrow$ v
$[_{\text{TP}}$ T $[_{\text{vP}}$ $\alpha_{[\tau]}$ $[_{\text{v}'}$ $\text{v}_{[\tau]}$ $[_{\text{TP}}$ $\text{t}''_{[\tau]}$ [$\text{T}_{[\tau]}$ $[_{\text{vP}}$ $\text{t}'_{[\tau]}$ $[_{\text{v}'}$ $\text{v}_{[\tau]}$ $\ldots$

(ii) John believed Mary_1 (sincerely) $[_{\text{TP}}$ t_1 to be the winner]

(iii) John believed $[_{\text{TP}}$ Mary $\text{t}_1]$ (sincerely) $[\text{to be the winner}]_1$

[27] One remaining question is whether features are associated with a default altitude. Throughout this paper, I maintain the view that the altitude of features is arbitrary and propose a model that does not constrain the altitude itself. As noted by Abels (2007), the UCOOL strikingly mirrors the height of the landing sites of various movement types, such that the hierarchy of the *fseq* constituting the clausal spine

8 Conclusion

The point of departure in this paper is an exploration of the consequences of successive cyclicity for accounts of improper movement, particularly those based on the Williams Cycle (WC). Given the successive cyclic nature of movement, there has been a debate on whether the WC should ignore intermediate movement. Cross-clausal movement in the *lian… dou* construction in Mandarin Chinese represents a crucial typological case that sheds light on this question. I showed that TP-internal movement in the *lian… dou* construction is clause-bound, while TP-external movement can cross CPs. Crucially, the distinction between licit and illicit cross-CP movement hinges on whether TP-internal movement is intermediate or terminal. This pattern suggests that intermediate movement steps must be alleviated from constraints on improper movement. I propose that the WC should be reformulated as an order in which triggering features must be interpreted along the movement path. Under this reformulation, a movement path is deemed improper if movement that interprets a given feature is extended by movement that interprets a lower-ranked feature.

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and that of the UCOOL are structurally isomorphic. I do not know whether this resemblance is coincidental or reflects an underlying principle linking the two. I leave this issue for future research.

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